Class 12 Biology

Study Notes

Chapter 1: Photosynthesis

Section 1.1: Light Reaction

The light reaction takes place in the thylakoid membranes of the chloroplast. Chlorophyll absorbs light, mainly in the red and blue wavelengths, and uses the energy to split water molecules in a process called photolysis. Photolysis releases oxygen as a by-product and supplies electrons to the electron transport chain. The energy captured is stored in two molecules, ATP and NADPH.

Section 1.2: Calvin Cycle

The Calvin cycle takes place in the stroma of the chloroplast and does not need light directly. It uses the ATP and NADPH produced in the light reaction. The enzyme RuBisCO fixes carbon dioxide onto a five-carbon sugar called RuBP. The resulting molecules are reduced to G3P, which is used to make glucose. Six turns of the cycle are needed to make one molecule of glucose.

Section 1.3: Limiting Factors

The rate of photosynthesis is controlled by the factor that is in shortest supply. Light intensity, carbon dioxide concentration and temperature are the main limiting factors. Raising light intensity increases the rate until another factor becomes limiting. Above the optimum temperature, enzymes such as RuBisCO begin to denature and the rate of photosynthesis falls sharply.

Chapter 2: Cellular Respiration

Section 2.1: Glycolysis

Glycolysis takes place in the cytoplasm and does not need oxygen. One glucose molecule with six carbons is split into two molecules of pyruvate with three carbons each. Two ATP are used up and four ATP are produced, so the net gain is two ATP. Two molecules of NADH are also formed.

Section 2.2: Krebs Cycle

The Krebs cycle takes place in the mitochondrial matrix. Pyruvate is first converted to acetyl CoA, which then enters the cycle. Each turn of the cycle produces three NADH, one $FADH_2$ and one ATP, and releases two molecules of carbon dioxide. The cycle turns twice for every glucose molecule.

Section 2.3: Electron Transport Chain

The electron transport chain is located on the inner mitochondrial membrane. NADH and $FADH_2$ donate electrons, which pass along protein complexes and pump protons into the intermembrane space. The protons flow back through the enzyme ATP synthase, which produces most of the ATP made in respiration. Oxygen is the final electron acceptor and combines with electrons and protons to form water. Without oxygen the chain stops.

Section 2.4: Anaerobic Respiration

When oxygen is not available, pyruvate is converted to lactic acid in animal muscle cells, or to ethanol and carbon dioxide in yeast. This regenerates NAD+ so that glycolysis can continue. However, anaerobic respiration gives only two ATP per glucose molecule, far less than aerobic respiration.

Chapter 3: Cell Division

Section 3.1: Cell Cycle and Checkpoints

The cell cycle consists of interphase and the mitotic phase. Interphase is divided into G1, S and G2, and DNA is replicated during the S phase. Checkpoints at G1, G2 and metaphase make sure that the DNA is undamaged and the chromosomes are correctly attached before the cell continues. Failure of these checkpoints can lead to uncontrolled division and cancer.

Section 3.2: Mitosis

Mitosis produces two genetically identical diploid daughter cells from one parent cell. Its stages are prophase, metaphase, anaphase and telophase. In metaphase the chromosomes line up at the equator of the cell, and in anaphase the sister chromatids are pulled apart. Mitosis is used for growth and for the repair of tissues.

Section 3.3: Meiosis

Meiosis involves two successive divisions and produces four genetically different haploid gametes. Homologous chromosomes separate in meiosis I, and sister chromatids separate in meiosis II. Crossing over occurs in prophase I, and independent assortment occurs in metaphase I. Both processes increase genetic variation.

Chapter 4: Genetics

Section 4.1: Mendel's Laws

The law of segregation states that the two alleles for a trait separate during gamete formation, so each gamete carries only one allele. The law of independent assortment states that genes for different traits are inherited independently of each other. The law of dominance states that in a heterozygous individual the dominant allele masks the recessive allele.

Section 4.2: Monohybrid Cross

A cross between two heterozygous parents (Tt x Tt) gives offspring TT, Tt, Tt and tt. The genotypic ratio is 1:2:1 and the phenotypic ratio is 3:1 for dominant to recessive. A test cross with a homozygous recessive parent shows whether an individual with the dominant trait is homozygous or heterozygous.

Section 4.3: DNA Structure and Replication

DNA is a double helix made of two antiparallel strands of nucleotides. Adenine pairs with thymine through two hydrogen bonds, and guanine pairs with cytosine through three hydrogen bonds. DNA replication is semi-conservative, because each new molecule contains one old strand and one new strand. The enzyme helicase unwinds the helix and DNA polymerase adds new nucleotides.

Chapter 5: Ecosystems

Section 5.1: Food Chains and Energy Flow

A food chain begins with producers, which are followed by primary consumers, secondary consumers and decomposers. Only about ten percent of the energy at one trophic level is passed to the next, which is known as the ten percent law. The rest is lost as heat through respiration. Because of this, food chains rarely have more than four or five trophic levels.

Section 5.2: Nutrient Cycles

In the carbon cycle, photosynthesis removes carbon dioxide from the air, while respiration and combustion return it. In the nitrogen cycle, nitrogen-fixing bacteria convert atmospheric nitrogen into ammonia. Nitrifying bacteria convert ammonia into nitrates, which plants absorb. Denitrifying bacteria return nitrogen gas to the atmosphere.

Section 5.3: Ecological Pyramids

Ecological pyramids show the relationship between trophic levels. There are pyramids of numbers, biomass and energy. The pyramid of energy is always upright because energy decreases at each level. A pyramid of numbers can be inverted, for example when one tree supports thousands of insects.